Huawei Access Node MAR-ACC-26: SOP for Resolving Optical Transport Failure (SFP/XFP Degradation, Fiber Cuts, Rx/Tx Power Loss) - Alarm Code: ALARM_9406

Node: MAR-ACC-26 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_9406

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Huawei Access Node MAR-ACC-26 at the Marathahalli Core site can be attributed to its role in handling Access traffic. The Optical Transport Network (OTN) equipment, specifically the SFP/XFP modules, plays a crucial role in transmitting data over optical fibers. When these modules degrade or experience fiber cuts, it can lead to a significant reduction in Receive (Rx) and Transmit (Tx) power levels, resulting in the ALARM_9406 alarm. This is due to the OTN's reliance on wavelength division multiplexing (WDM) and the inherent characteristics of optical signals, which make them susceptible to attenuation, dispersion, and other impairments. Furthermore, the Huawei equipment's configuration and protocol stack, such as the OTN's G.709 protocol, also contribute to the likelihood of this failure domain. In particular, the Huawei OTN's use of the G.709 protocol for Forward Error Correction (FEC) and its reliance on the SFP/XFP modules for signal regeneration and amplification increase the vulnerability of the network to optical transport failures.

Pre-requisite Checks

- pre-condition: Verify that the site is accessible for physical interventions., task: Dispatch a technician to the Marathahalli Core site to investigate the issue.

- pre-condition: Check the status of redundant links in the network., task: Verify that the redundant links are operational and can take over traffic in case of a failure.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss Due to OTN Impairments	Low

Diagnostic Commands & Expected Outputs

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Command: display optical-transport information, Expected Output: OTN information, including SFP/XFP module status and optical signal quality, Description: This command displays the OTN information, including the SFP/XFP module status and optical signal quality, which
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can help identify the root cause of the issue.

Command: display optical-transport sfp-xfp, Expected Output: SFP/XFP module information, including temperature, voltage, and optical power levels, Description: This command displays the SFP/XFP module information, including temperature, voltage, and optical power levels, which can help determine if the module is functioning correctly.

Command: display optical-transport alarm, Expected Output: Alarm information, including the ALARM_9406 alarm, Description: This command displays the alarm information, including the ALARM_9406 alarm, which can help confirm the issue.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP Module, Command: replace optical-transport sfp-xfp {module-id}, Description: This step involves replacing the SFP/XFP module with a new one. The {module-id} should be replaced with the actual module ID of the faulty module.

Step 2: Step: Verify the Optical Signal Quality, Command: display optical-transport information, Description: This step involves verifying the optical signal quality after replacing the SFP/XFP module to ensure that the issue is resolved.

Step 3: Step: Configure the OTN to Use the New SFP/XFP Module, Command: config optical-transport sfp-xfp {module-id} {interface-name}, Description: This step involves configuring the OTN to use the new SFP/XFP module. The {module-id} should be replaced with the actual module ID of the new module, and the {interface-name} should be replaced with the actual interface name.

Post-Check Verification

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Command: display optical-transport information, Expected Output: OTN information, including SFP/XFP module status and optical signal quality, Description: This command displays the OTN information, including the SFP/XFP module status and optical signal quality, which can help verify that the issue is resolved.

Command: display optical-transport sfp-xfp, Expected Output: SFP/XFP module information, including temperature, voltage, and optical power levels, Description: This command displays the SFP/XFP module information, including temperature, voltage, and optical power levels, which can help verify that the SFP/XFP module is functioning correctly.

Command: display optical-transport alarm, Expected Output: Alarm information, including the absence of the ALARM_9406 alarm, Description: This command displays the alarm information, including the absence of the ALARM_9406 alarm, which can help confirm that the issue is resolved.
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Roll-back Procedures

- Step: Restore the Original SFP/XFP Module, Command: replace optical-transport sfp-xfp {module-id} revert, Description: This step involves restoring the original SFP/XFP module. The {module-id} should be replaced with the actual module ID of the original module.

- Step: Re-configure the OTN to Use the Original SFP/XFP Module, Command: config optical-transport sfp-xfp {module-id} {interface-name} revert, Description: This step involves re-configuring the OTN to use the original SFP/XFP module. The {module-id} should be replaced with the actual module ID of the original module, and the {interface-name} should be replaced with the actual interface name.

- Step: Verify the Original SFP/XFP Module, Command: display optical-transport information, Description: This step involves verifying the original SFP/XFP module to ensure that it is functioning

correctly.